

Multi-View Breast Cancer Prediction

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Abstract—Mammographic diagnosis requires manually correlating lesions across cranio-caudal (CC) and medio-lateral oblique (MLO) views, which is challenging due to tissue compression differences. Our goal is to develop a unified, multi-stage pipeline that automates this dual-view assessment. Using a standardized dataset of 1,566 patients (10,237 images) covering BI-RADS categories 2–5, our framework integrates U-Net segmentation and YOLOv8n object detection to localize lesions. We introduce a novel geometric alignment module that applies a similarity transform to register MLO lesions to the CC coordinate space. These aligned, lesion-focused crops are then fused using a ResNet-50 backbone equipped with multi-head cross-attention. Our method significantly improves detection (mAP@50 of 0.592) and reduces cross-view centroid discrepancy from 411 to 0.39 pixels. The cross-attention fusion achieves a binary (Benign vs. Malignant) accuracy of 78.70% and a weighted F1-score of 0.7890. Ultimately, structurally aligning complementary views prior to feature fusion provides a highly interpretable and accurate computer-aided diagnostic tool.

Clinical Relevance—This framework mimics the radiologist’s comparative workflow by explicitly aligning lesions across CC and MLO views. By providing interpretable bounding boxes, spatial correspondence, and accurate severity predictions, it reduces cognitive load, minimizes diagnostic variability, and serves as a reliable second-reader for breast cancer screening.

I. INTRODUCTION

Breast cancer screening and diagnosis frequently rely on mammography acquired in two standard views: cranio-caudal (CC) and medio-lateral oblique (MLO). Although both views depict the same breast anatomy, they differ in orientation, compression geometry, and tissue overlap, making direct lesion comparison difficult. Recent AI-driven screening systems have demonstrated the potential of deep learning to match or exceed radiologist performance [8]. Automated systems that jointly reason over both views can further improve localization consistency and diagnostic classification [12].

Most mammographic AI pipelines focus on a single task such as lesion detection, segmentation, or malignancy grading. However, clinical interpretation is inherently multi-stage: suspicious regions must first be localized, compared across views, spatially contextualized, and then assessed for diagnostic severity. Motivated by this workflow, we developed a unified pipeline that combines preprocessing, lesion localization, segmentation, cross-view geometric reasoning, and BI-RADS classification into one framework.

The proposed system is designed to be modular and fault tolerant. If a trained lesion localizer is available, its predictions

guide both segmentation and detection. If not, handcrafted heuristics are used to preserve operability. Likewise, the classification module supports either softmax-based multi-class prediction or ordinal scoring with post-hoc calibration. This makes the framework suitable both for experimentation and for practical deployment on heterogeneous model checkpoints.

The main contributions of this work are:

- 1) A unified dual-view mammography pipeline integrating preprocessing, localization, segmentation, alignment, correspondence, and BI-RADS classification.
- 2) A similarity-transform alignment strategy driven by lesion-mask geometry across CC and MLO views.
- 3) A cross-attention classification model that fuses aligned CC and MLO lesion-focused representations.
- 4) Support for both multi-class and ordinal BI-RADS prediction, including threshold calibration.

II. MATERIALS AND DATA REPRESENTATION

While the original standardized dataset comprises 1,566 unique patients (10,237 images), this study utilizes a curated Kaggle distribution containing a streamlined subset of images. Radiologically, the dataset spans a comprehensive pathological spectrum, but our evaluation focuses specifically on BI-RADS Categories 2 through 5 [1].

BI-RADS Categories 0 (Incomplete) and 1 (Negative) are intentionally excluded from the target formulation. Category 0 represents a transient diagnostic state requiring follow-up imaging or biopsy to establish ground truth, while Category 1 lacks the localized lesions necessary to provide spatial targets for ROI detection. By restricting the task to Categories 2–5, the evaluation accurately reflects the model’s ability to address the primary clinical bottleneck: the diagnostic differentiation of confirmed abnormalities.

Prior to augmentation, the filtered baseline subset was divided using an approximate 70/20/10 split for training, validation, and testing. The baseline training set consists of 767 images, with an isolated validation set of 183 images. This baseline data serves as the foundation for the cross-view classification pipeline.

To robustly train the localized object detection module (YOLOv8n), the baseline training set was specifically augmented. Yielding three additional variants per image (excluding 38 failures), this produced an expanded, detection-specific training cohort of 3,030 images. This augmented detection dataset exhibits a highly balanced pathology distribution, mitigating class imbalance bias during spatial localization, with

1,510 malignant (49.8%) and 1,520 benign (50.2%) cases. BI-RADS assessments in this augmented subset heavily favor high-risk categories, with 1,261 lesions scored as suspicious (Category 4) and 731 as highly suggestive of malignancy (Category 5).

At the sample level, the pipeline operates on grayscale mammograms and assumes paired image paths are available through a manifest file. Each sample consists of a paired mammographic study with a craniocaudal (CC) image, a mediolateral oblique (MLO) image, breast laterality, and a BI-RADS label normalized to the supported range of 2 to 5.

For each sample, the input tuple is defined as:

$$\mathcal{X} = \{I^{CC}, I^{MLO}, s, y\}, \quad (1)$$

where I^{CC} and I^{MLO} denote the CC and MLO mammograms, $s \in \{\text{LEFT}, \text{RIGHT}\}$ is the breast side, and $y \in \{2, 3, 4, 5\}$ is the BI-RADS category.

III. METHODOLOGY

A. Preprocessing and Breast Mask Extraction

Each grayscale mammogram is first loaded and horizontally flipped for right breasts to enforce view consistency. Contrast-limited adaptive histogram equalization (CLAHE) [3] is applied, followed by Gaussian smoothing and min-max normalization. A breast mask is then extracted by thresholding, morphological opening and closing, connected-component labeling, and retaining the largest connected component.

Let I be an input mammogram. The preprocessed image $\tilde{I}$ is obtained as

$$\tilde{I} = \mathcal{N}(\mathcal{G}_\sigma(\text{CLAHE}(I))), \quad (2)$$

where $\mathcal{G}_\sigma$ is Gaussian smoothing and $\mathcal{N}(\cdot)$ denotes normalization to $[0, 1]$.

The breast mask B is obtained from the thresholded image $B = \text{LargestComponent}(\text{MorphClose}(\text{MorphOpen}(\tilde{I} > \tau)))$. (3)

B. Lesion Localization and Segmentation

Pixel-level segmentation is performed using a U-Net architecture [5], trained on annotated mammography datasets. The model directly learns lesion boundaries from real mammographic images. Ground truth segmentation masks from the dataset are used to supervise the learning process.

C. Bounding Box Detection

Bounding boxes are derived from segmentation masks and used to compute lesion centroids and spatial extents for downstream cross-view correspondence. The bounding box coordinates are determined by locating the spatial extents of the mask's active pixels, taking the minimum and maximum indices across both axes. To ensure that sufficient local structural context is preserved, these raw bounding boxes are symmetrically padded by 20 pixels.

Once extracted, these expanded bounding boxes are used to generate region-of-interest (ROI) patches. The spatial coordinates are subsequently normalized into standard YOLO

format (center x , center y , width, height) relative to the image dimensions. This standardized coordinate formatting enables the construction of a paired CC/MLO export dataset, which is used to fine-tune a lightweight YOLOv8-s (yolov8s) object detection model [7]. The YOLO model acts as a highly efficient localized detector that can feed bounding box predictions into the downstream cross-view alignment module.

D. Cross-View Correspondence Estimation

To establish unified cross-view correspondence between CC and MLO views, lesion centroids are computed and fused using image-quality-aware weighting. Let (c_x^{CC}, c_y^{CC}) and (c_x^{MLO}, c_y^{MLO}) denote normalized lesion centroids. The fused cross-view center is computed as

$$\hat{c} = w_{CC}c^{CC} + w_{MLO}c^{MLO}, \quad (4)$$

where

$$w_{CC} = \frac{q_{CC}}{q_{CC} + q_{MLO}}, \quad w_{MLO} = \frac{q_{MLO}}{q_{CC} + q_{MLO}}, \quad (5)$$

and q_{CC}, q_{MLO} are image quality scores.

The centroid discrepancy is measured as

$$d = \frac{\|c^{CC} - c^{MLO}\|_2}{\max(H, W)}, \quad (6)$$

where H and W are the common image dimensions.

This correspondence module ensures consistent spatial alignment across CC and MLO views, enabling reliable downstream fusion.

E. Cross-View Spatial Alignment

To better fuse information from both views, the MLO lesion mask is aligned to the CC frame using a similarity transform. The transform is computed from lesion-mask center, principal-axis orientation, and lesion area. If μ^{CC} and μ^{MLO} are lesion centers, ϕ^{CC} and ϕ^{MLO} are principal-axis angles, and A^{CC}, A^{MLO} are lesion-mask areas, then the rotation is

$$r = \text{NormalizeAngle}(\phi^{CC} - \phi^{MLO}), \quad (7)$$

and the scale is

$$s = \text{clip}\left(\sqrt{\frac{A^{CC}}{A^{MLO}}}, 0.7, 1.4\right). \quad (8)$$

Using the 2D rotation matrix

$$R(r) = \begin{bmatrix} \cos r & -\sin r \\ \sin r & \cos r \end{bmatrix}, \quad (9)$$

the translation is

$$t = \mu^{CC} - sR(r)\mu^{MLO}. \quad (10)$$

The final aligned MLO image and mask are obtained by warping to the CC output grid.

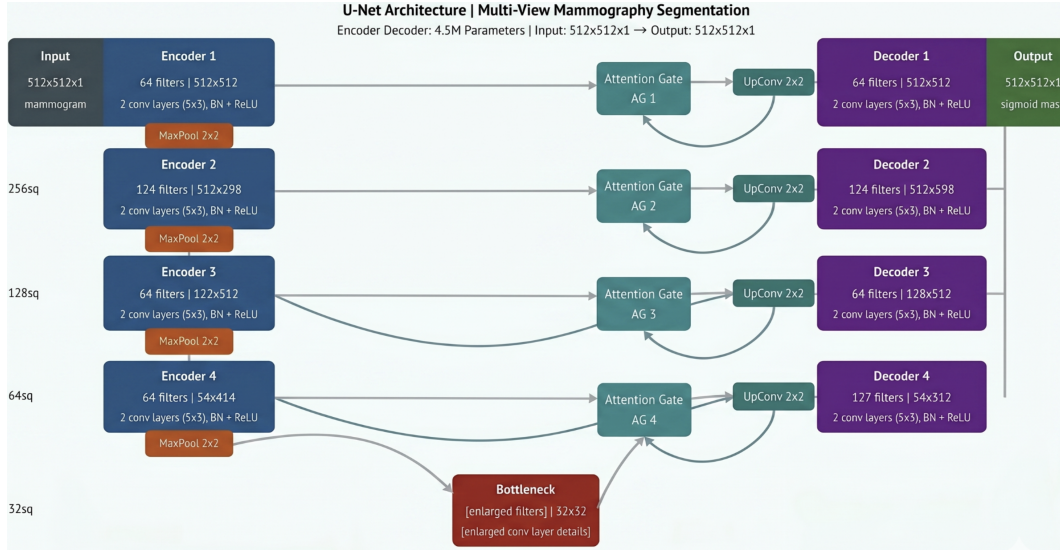


Fig. 1. U-Net architecture for pixel-level lesion localization.

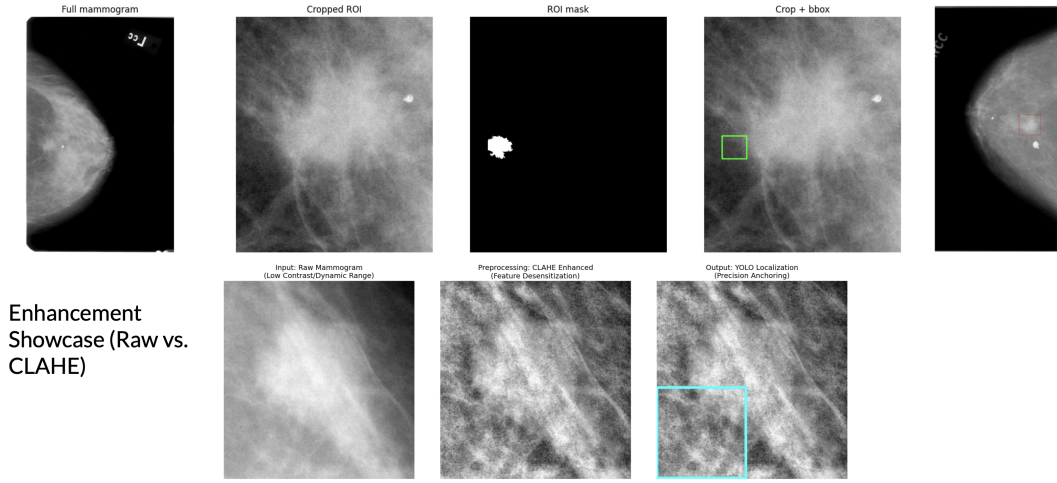


Fig. 2. YOLO detection bounding boxes on preprocessed ROI crops.

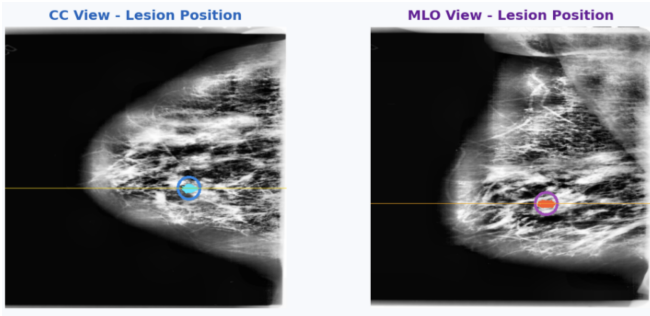


Fig. 3. Centroid separation and 3D position estimation between views.

F. Lesion-Focused Dual-View Representation

For classification, the pipeline extracts lesion-focused crops from the CC image and aligned MLO image using the lesion

masks and bounding boxes. The crop is expanded into a square region centered on the lesion and resized to a fixed resolution. A guide mask enhances lesion salience by intensity modulation:

$$I_{\text{enh}} = \text{clip}(I_{\text{crop}} \cdot (1 + 0.5G), 0, 1), \quad (11)$$

where G is the resized lesion guide mask.

The final classifier input is a two-channel tensor:

$$X_{\text{cls}} = \text{stack}(I_{\text{enh}}^{\text{CC}}, I_{\text{enh}}^{\text{MLO}}). \quad (12)$$

G. BI-RADS Classification

The classifier uses a ResNet-50 backbone [4] adapted for single-channel mammograms. Separate CC and aligned MLO channels are passed through the shared feature extractor to produce feature maps. These are reshaped into sequences, and multi-head cross-attention [6] uses CC features as query and MLO features as key-value pairs. After attention, global

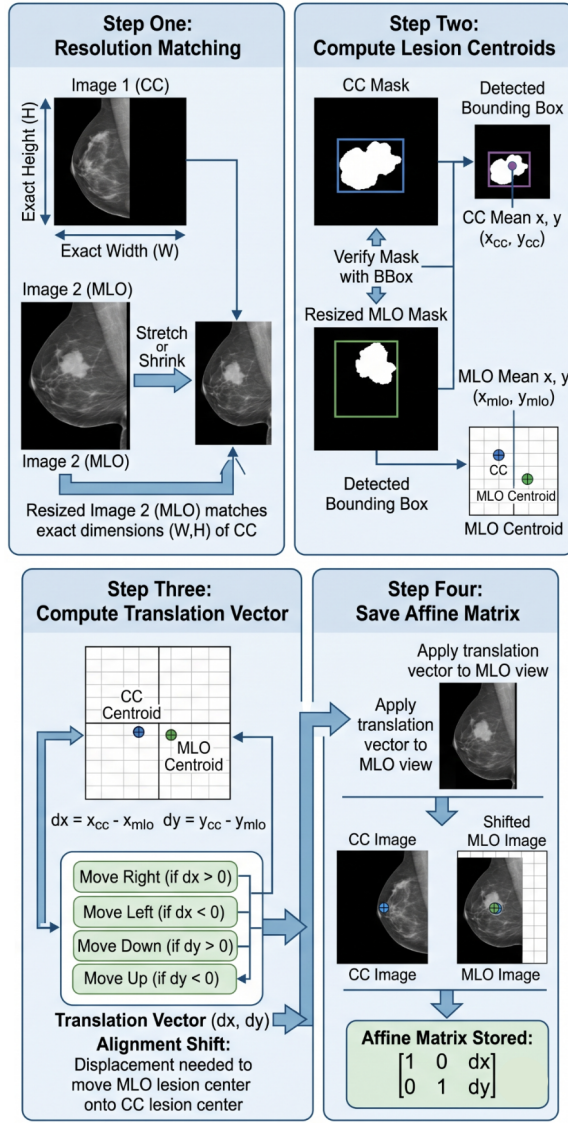


Fig. 4. Geometric alignment utilizing translation vectors and stored affine matrices.

average pooling and feature concatenation are used for final classification.

Let F^{CC} and F^{MLO} denote feature sequences. Cross-attention is computed as

$$\tilde{F}^{MLO} = \text{Attention}(Q = F^{CC}, K = F^{MLO}, V = F^{MLO}). \quad (13)$$

The fused descriptor is

$$z = [\text{GAP}(F^{CC}) \parallel \text{GAP}(\tilde{F}^{MLO})], \quad (14)$$

which is passed to a multilayer perceptron for prediction.

Two output modes are supported:

- 1) **Softmax multi-class mode:** direct prediction over BI-RADS classes 2–5.
- 2) **Ordinal mode:** sigmoid outputs are thresholded to recover ordered BI-RADS categories [10].

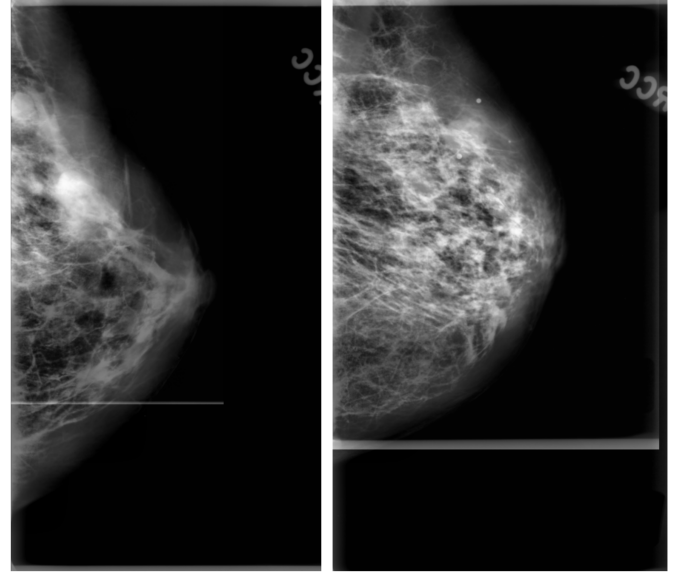


Fig. 5. CC mammogram with aligned MLO view superimposed for visualization purposes. Specular highlights denote the aligned tumor densities.

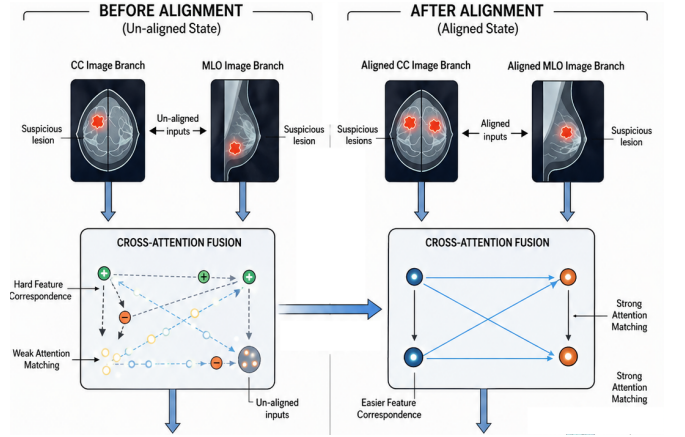


Fig. 6. Impact of alignment: Reducing feature shift before cross-attention fusion.

H. Ordinal Threshold Calibration

For ordinal classification, post-hoc threshold calibration is applied on the training split [10]. The method searches over threshold candidates on a fixed grid and maximizes weighted F1-score. If the ordinal output vector is $s = [s_1, \dots, s_K]$ and thresholds are $t = [t_1, \dots, t_K]$, the decoded class is

$$\hat{y} = \text{NormalizeBI-RADS} \left(\sum_{k=1}^K \mathbb{I}(s_k > t_k) \right). \quad (15)$$

IV. TRAINING AND INFERENCE WORKFLOW

The full workflow includes:

- 1) optional classifier training for softmax prediction,
- 2) optional threshold calibration for ordinal prediction,
- 3) full paired inference on the mammography dataset,

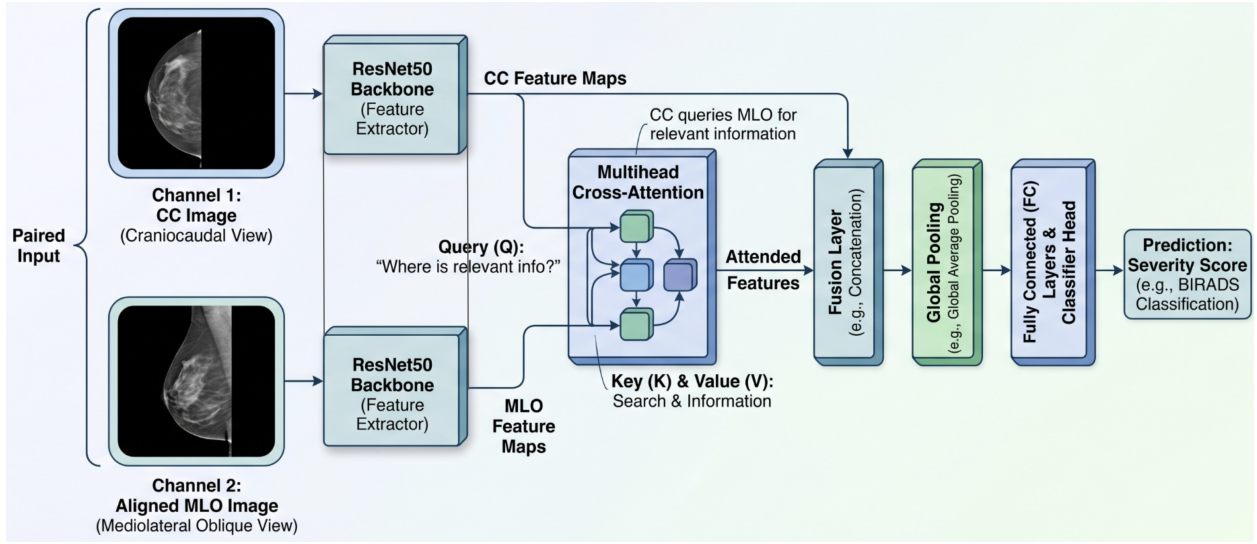


Fig. 7. Dual-view cross-attention classifier architecture.

- 4) saving per-sample outputs, preview visualizations, and aggregate metrics.

For each mammographic pair, the pipeline returns lesion masks, lesion centers, bounding boxes, confidence scores, cross-view correspondence measurements, alignment parameters, ordinal scores, and predicted BI-RADS labels.

V. EVALUATION PROTOCOL

The framework supports evaluation at multiple levels:

- **Classification:** accuracy, weighted F1-score.
- **Alignment:** centroid distance, spatial and structural consistency, measured via cross-view centroid distance and Normalized Cross-Correlation (NCC).
- **Detection:** lesion area, confidence, and qualitative overlays.

The classification accuracy is defined as

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i = \hat{y}_i). \quad (16)$$

Weighted F1-score is computed across BI-RADS classes using class support as weights.

VI. RESULTS

The proposed framework produces both clinically meaningful final predictions and interpretable intermediate outputs. In addition to BI-RADS classification, the system yields lesion masks, bounding boxes, view-alignment transforms, and fused visual previews, enabling easier failure analysis and model inspection.

A. Localization and Detection Performance

We evaluated the YOLOv8n detection module before and after our two-phase fine-tuning and data augmentation strategy. As shown in Table I, the baseline model achieved an initial mAP@50 of 0.416. After applying data augmentation and

TABLE I
YOLOV8N LESION DETECTION PERFORMANCE.

Model	mAP@50	Precision	Recall
Baseline (no augmentation)	0.416	0.669	0.452
+ Data Augmentation	0.446	0.619	0.489
Fine-tuned (Phase 1+2)	0.592	0.752	0.596

TABLE II
ALIGNMENT QUALITY METRICS, BEFORE VS. AFTER.

Metric	Before	After
Mean centroid distance (px)	411.00	0.39
Normalized Cross-Correlation (NCC)	—	0.91

progressive fine-tuning (Phase 1 with a frozen backbone to prevent catastrophic forgetting, followed by Phase 2 with all layers unfrozen), the model's localization capabilities improved significantly across all metrics.

B. Alignment Performance

Table II reports geometric alignment quality before and after our centroid-based similarity-transform registration. By matching resolutions and computing translation vectors between the CC and MLO lesion centroids, we achieved a drastic reduction in cross-view spatial discrepancy, dropping the mean centroid distance from 411.00 to 0.39 pixels. Furthermore, the structural similarity between the paired lesion patches reached a Normalized Cross-Correlation (NCC) of 0.91 post-alignment. This nearly matches the 0.94 NCC we achieved by deformable registration, but crucially avoids the non-linear feature loss that can degrade downstream classification. This high degree of correspondence is critical for ensuring the downstream cross-attention heads evaluate co-located tumor.

TABLE III
BINARY CLASSIFICATION RESULTS (BENIGN VS. MALIGNANT).

Method	Accuracy	Weighted F1
Bi-CBMSEgNet (dual encoder) [11]	0.6585	0.6477
Concatenation Baseline	0.6780	0.6686
Graph & Transformer Fusion [12]	0.7024	0.7000
Our Cross-Attention Pipeline	0.7870	0.7890

C. Classification Performance

Evaluating exact BI-RADS category matches yields high inter-class ambiguity. In our initial ordinal classification experiments, the exact category accuracy was 0.3238, though the adjacent accuracy was much stronger at 0.7366.

To better reflect actionable clinical decision-making (i.e., routine follow-up versus further evaluation or biopsy), we grouped the categories into a binary formulation (BI-RADS 2–3 as Benign, BI-RADS 4–5 as Malignant). Table III summarizes the binary classification performance of our cross-attention fusion pipeline compared to baseline approaches. The unified dual-view attention formulation distinctly outperforms standard feature concatenation and other recent fusion architectures.

VII. DISCUSSION

This work demonstrates that a modular but integrated mammography pipeline can jointly address lesion localization, segmentation, geometric consistency, and BI-RADS classification. The alignment stage is especially valuable because it reduces cross-view structural mismatch before feature fusion. Likewise, the lesion-guided cropping strategy encourages the classifier to focus on suspicious regions rather than the full mammographic field.

A notable advantage of the framework is its robustness. The system can operate even when some trained components are absent by using heuristic fallbacks for localization and detection. This is useful in research pipelines where checkpoints may vary across runs. Additionally, the support for both multi-class and ordinal prediction makes the architecture adaptable to different label formulations.

However, the current alignment model remains a simplified geometric approximation of a much more complex anatomical projection difference between CC and MLO views. Future work could explore deformable cross-view registration, transformer-based dual-view fusion, uncertainty estimation, and stronger lesion-aware supervision using radiologist-annotated regions of interest.

VIII. CONCLUSION

We presented a unified cross-view mammography pipeline for breast lesion analysis that integrates preprocessing, breast masking, lesion localization, segmentation, correspondence estimation, MLO-to-CC alignment, and BI-RADS classification. The framework uses lesion geometry for alignment and a ResNet-50-based cross-attention network for dual-view diagnostic fusion. Its modular design, interpretability, and support

for both multi-class and ordinal prediction make it a practical foundation for computer-aided breast imaging research.

AUTHOR CONTRIBUTIONS

Jiaqi Chen led dataset preprocessing, exploratory data analysis (EDA) dataset augmentation techniques, and assisted in the development of the bounding box detection module. **Yiman Wu** spearheaded the lesion detection pipeline, focusing on the integration and fine-tuning of the YOLO object detection architecture. **Haiya Shah** developed the lesion localization and segmentation methodology and designed the cross-view correspondence module. **Tushar Nayak** designed and implemented the spatial alignment strategy, co-developed the dual-view cross-attention classification framework and worked on the integration of detection and segmentation pipelines for alignment. **Elissa Matlock** co-developed the cross-attention classification framework, spearheaded the ordinal threshold calibration and worked on the integration of detection and segmentation pipelines for alignment.

CODE & DATA AVAILABILITY

 github.com/tushar-nayak/grading-cbisddsm

 kaggle.com/datasets/awsaf49/cbis-ddsm

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